Moonshot versus Incremental: Innovation, Competition and Product Ranges

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Innovation in an Uncertain World

When firms innovate, there are considerable risks:

- Need to commit to innovation strategy before outcome is clear
- Innovation may fail
- Targeting the precise consumer needs a priori is hard
- A competitor may also innovate in that area

This paper: Taking the "innovate first $\Rightarrow$ adjust later" setting seriously

Central modeling choice: Firms competitively decide how to innovate, then adjust once uncertainty is lifted.

Example: Apples biggest failure

Apple Newton

Apple decides that Handhelds are the future $\rightarrow$ start developing the Newton

idea: aiming to build a smartphone with 1980ies tech

massive flop: too clunky, too heavy, to complicated

consequences: large resources go into adjusting to consumer feedback and demand

PalmPilot

Palm decides to develop efficient writing on touch screens

idea: use relatively low-fi mobile computing technology to supply a notebook

massive hit: much cheaper and directly catering to consumers' needs.

Different Strategies, Different Outcomes

Apple build a technology that could cater a wide range. But adjustment to individual consumer costly (lots of dependency, redundancy, . . .).
Palm's technology much less groundbreaking, but adjustment cost only small $\Rightarrow$ cheaper product (product already close to the needs).

Tradeoffs:
Tradeoff 1 (adjustment): More range, or more precision?
Tradeoff 2 (competition): Contesting the other's market, or focusing on a niche?

Research Question: In a competitive innovative market, how do firms solve these tradeoffs?

Related problems: Team Composition, Product Portfolio, . . .

Model

- Two (ex-ante identical) firms, A and B, both producing base good 0.
- Both independently decide on innovating in one product $x_i \in \mathbb{R}_0^+$ (at no cost).

  Convention: $x_A \geq x_B$
- One market $x \in [0, X]$ realizes at random.
- Firms use knowledge of *own* x_i (if successful) to offer x to the consumer at a chosen price p_i suffering adjustment costs $c(x; x_i)$ to produce the desired product $y(x)$ (next slide).
- The consumer has value v and unit demand for the “right” product $y(x)$, and decides (after observing prices) where to buy it (if at all).
- Payoffs realize.

Innovation & Adjustment Cost

We assume there is an order in the market, such that the "right" product for market x is derived via a Wiener process $y(x)$.

Innovation in x_i reveals $y(x_i)$ (and $y(x_i)$ only).

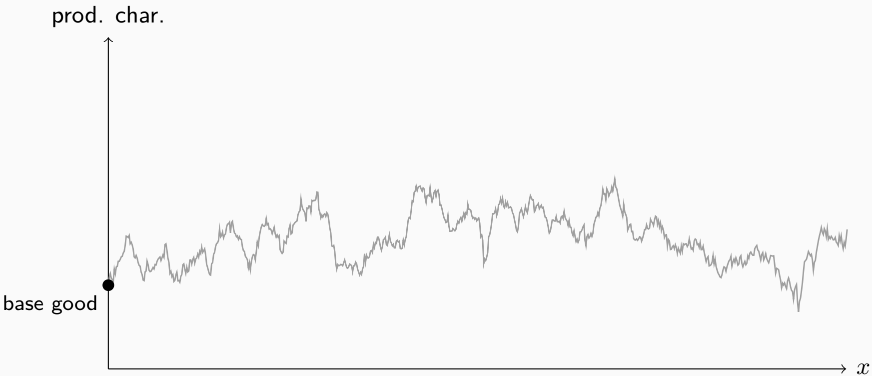
If market x realizes, adjustment cost are the cost "to fit" the product to market x.

We assume the realized cost are $(y(x) - \mu(x))^2$ where $\mu(x) = xy(x_i)$ for $x < x_i$ and $y(x_i)$ otherwise. Thus, in expectations:

$$c(x; x_i) := \begin{cases} \frac{x(x_i - x)}{x_i} & \text{if } x \leq x_i \\ \delta(x - x_i) & \text{if } x > x_i. \end{cases}$$

Parameter $\delta \geq 1$ captures an outside-the-range penalty.

Where the Adjustment Cost Comes From



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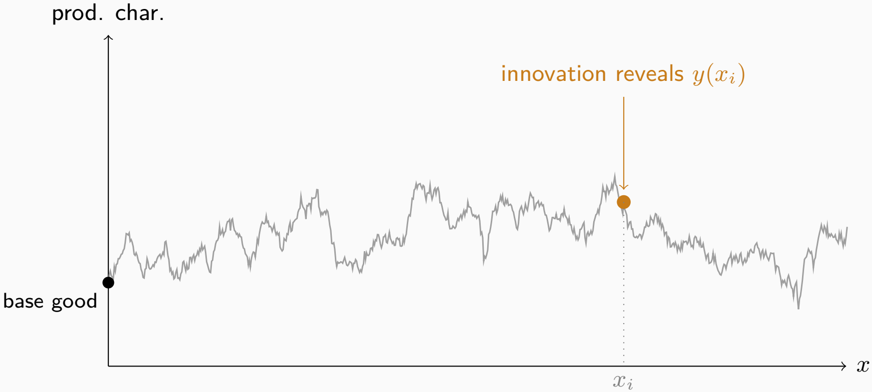

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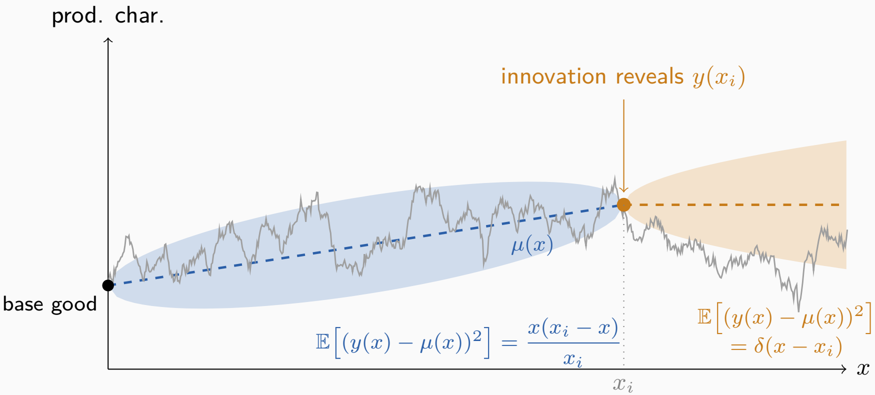

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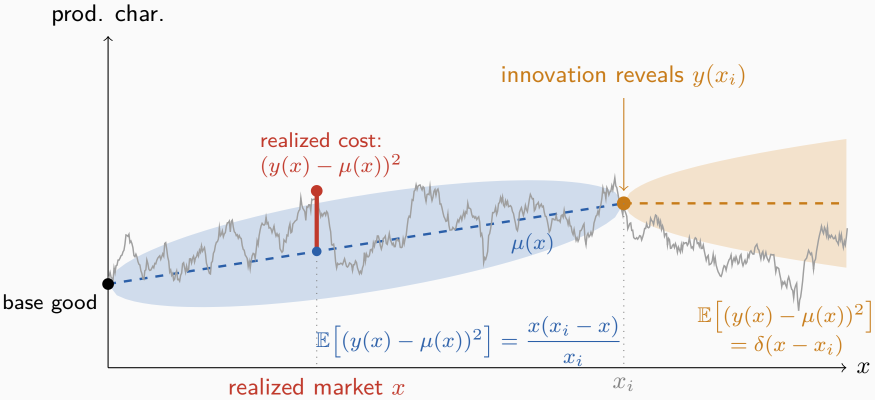
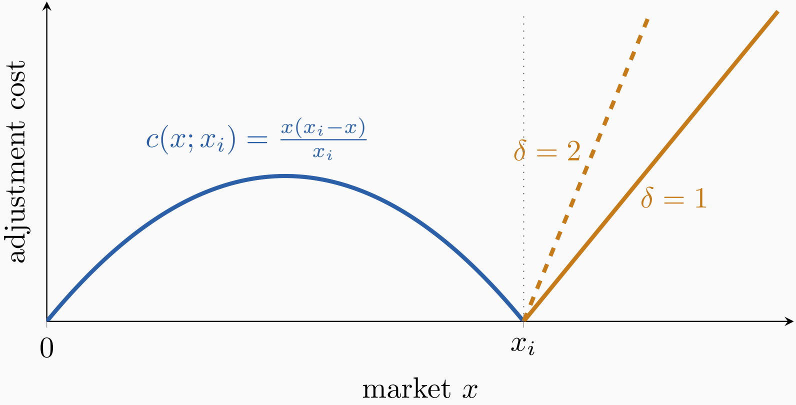


Adjustment Cost: Illustration



Warm-Up: Monopoly

The monopolist still needs to solve the range/precision tradeoff

$$\max_{x_i} \int_0^{x_i} \max\{v - c(x; x_i), 0\} dx + \int_{x_i}^{x_i + v/\delta} v - \delta(x - x_i) dx$$

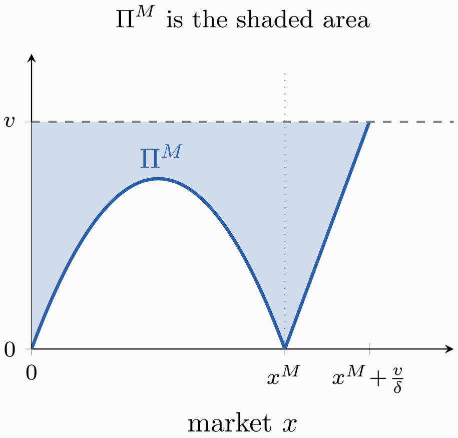
The second term is constant, so we only need to maximize the first.

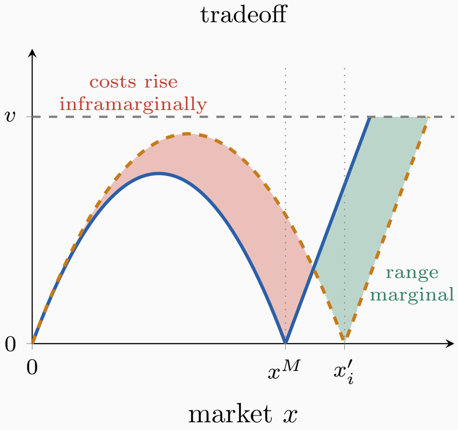
Proposition

The monopoly innovation choice is $x^M = 3v$.

Sidenote: For the optimization problem success probability is irrelevant (also because innovation is costless)

The Monopolist's Trade-off





Bring in the Competitor

Assume innovation is deterministic (always succeeds) and there are the two firms.

A is the avantgarde firm, $x_A \geq x_B$.

- Never makes sales on markets $x \leq x_B$
- sells on markets $x > x_B$,
- is quasi monopolist on markets $x > x_B + v/\delta$

Firm A's problem is

$$\begin{aligned}\max_{x_A} \int_{x_B}^{x_B+v/\delta} & \max\{\delta(x - x_B) - c(x; x_A), 0\} dx \\ &+ \int_{x_B+v/\delta}^{x_A} \max\{v - c(x; x_A), 0\} \\ &+ \int_{x_A}^{x_A+v/\delta} v - \delta(x - x_A) dx.\end{aligned}$$

The Other Firm

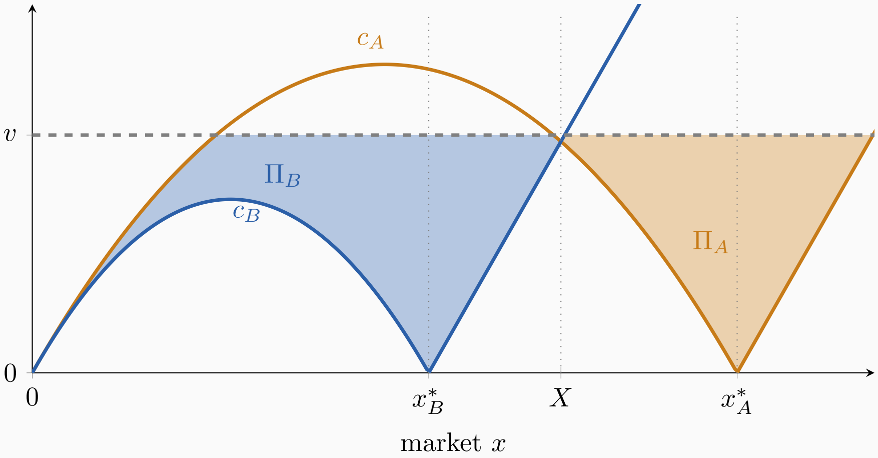
Firm B is the blue-chip firm.

- Never makes sales outside $x \in [0, x_B]$
- is contested by A at least in some range.

Firm B's problem is

$$\max_{x_B} \int_0^{x_B} \max \Big\{ \min\{c(x; x_A), v\} - c(x; x_b), 0 \Big\} dx$$

Oligopoly Profits



Housekeeping

Lemma

No Symmetric Equilibrium exists.

Lemma

Innovation is bounded even if $X \to \infty$.

Lemma

Ignoring the bound on X, for any x_i chosen by firm i, there exists an interior best response $x_j \in [0, \infty)$ by firm j.

Equilibrium

Assume X large. Everything here can be solved in closed form:

Proposition

In equilibrium

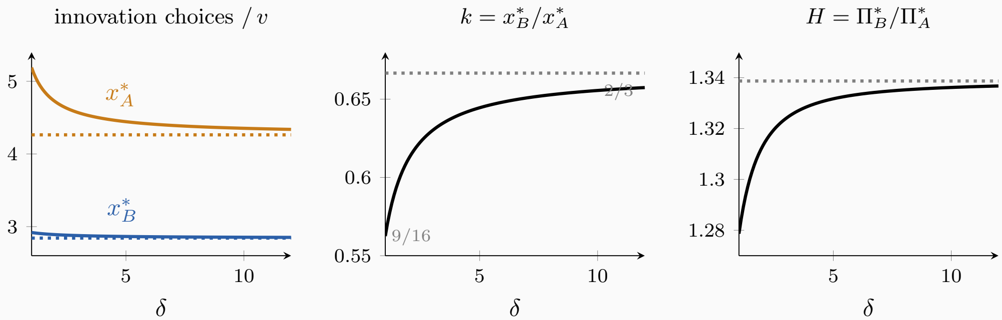
$$x_A^* = \frac{3(3\delta+1)^2}{\delta(19\delta^2+15\delta+3)}v, \quad x_B^* = \frac{9(2\delta+1)(3\delta+1)}{19\delta^2+15\delta+3}v.$$

In the equilibrium $H := \Pi_B^*/\Pi*_A =$ and $k := x_B/x_A \in [9/16, 2/3]$.

Corollary

Individual equilibrium profits $\Pi_i(\delta)$ decline in δ, and both $H(\delta)$ and $k(\delta)$ are monotone increasing in δ.

Comparative Statics in δ



Introducing Uncertainty

Now assume innovation is uncertain and may fail.

$\beta \in [0,1]$: Prob. both firms succeed

$\gamma \in [0,1/2]$: Prob. of one sided success (assumed symmetry)

$1-\beta-\gamma$: Prob. neither succeeds

Measure correlation as $\rho := \frac{\beta}{\beta+\gamma/2}$

If they correlate how does it change equilibrium behavior

Note: Innovation is costless, so correlation captures everything we need.

Housekeeping

Lemma

Perfect negative correlations $\rho = -1$ implies that behavior is like in the monopoly case.

Perfect positive correlation $\rho = 1$ implies it is like in the competitive case analysed before.

Across the correlation spectrum

Proposition

Fix $\delta \geq 1$ and $\rho \in [-1, 1]$.

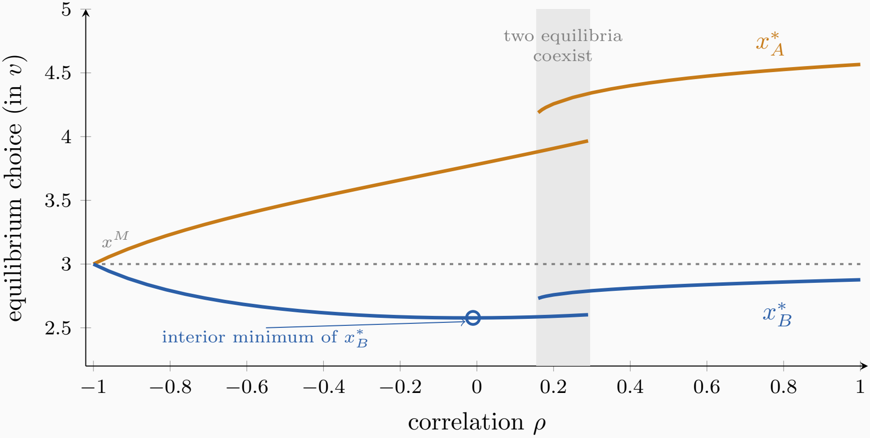
1. An equilibrium with $x_A > x_B$ exists.
2. In that equilibrium $\Pi_B > \Pi_A$

Proposition

Fix $\delta \geq 1$.
The minimum $x_B^*(\rho)$ is neither at $\rho = -1$, nor at $\rho = 1$.
The minimum $x_A^*(\rho)$ is at $\rho = -1$, the maximum at $\rho = 1$.

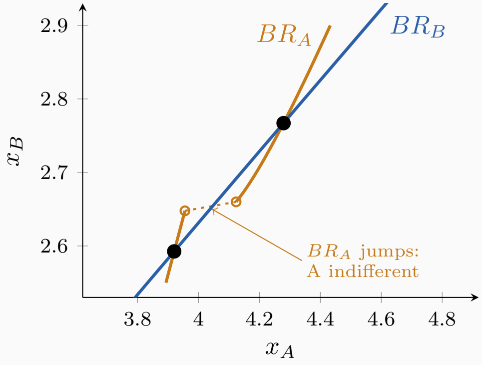
What does not carry over: Profit share H is not monotone in δ for all ρ (can have interior optimum)

The Correlation Spectrum in One Picture

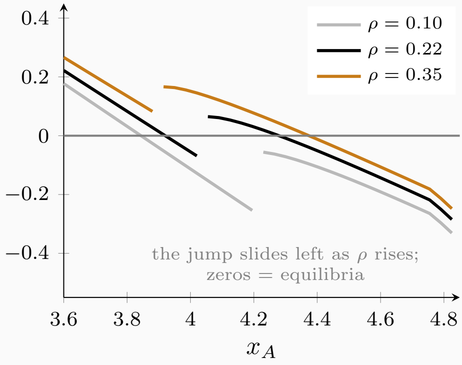


On the Coexistence

$\rho = 0.22$: best responses cross twice



$\varphi(x_A) - x_A, \ \varphi = BR_A \circ BR_B$



What else do we have

More general cost functions: $c(x;x_i) = x_i\psi\left(\frac{x}{x_i}\right)$ for $\psi : [0,1] \to \mathbb{R}^+$ concave, symmteric around $1/2$ and $\psi(0) = \psi(1) = 0$ in the interior.

Includes triangular cost and constant costs as limiting cases.

Sequential Innovation: One firm moves first (typically blue chip) the other moves later

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Conclusion

- Provide a model of competing firms in an innovative industry, that take innovation decisions expecting innovation to be
  - informative in meeting future demand
  - but not perfect $\Rightarrow$ adjustment cost looming
- Firms decide to differentiate more under competition leading to more innovation
- The blue-chip firm gains higher profits
- The blue-chip firms' innovation is non-monotone in the correlation parameter
- Dispersion is monotone in the correlation parameter
- If ex-post innovation becomes more cumbersome, the total span of innovation *decreases*